MTH205: ANALYSIS IN ONE VARIABLE

MidSem2 (Spring Semester 2025), IISER Mohali

SET E

Timing: 12 pm to 1 pm Date: March 06, 2025 Max. Marks: 20

Name: Reg. No:

1. Please answer all questions. The answer must have a justification/explanation otherwise, zero marks will be given even though the answer is correct.

2. Your name and registration number should be written on the question paper and answer sheet. At the time of submission, the question paper must be attached.

1. Prove or disprove (by counter example) the following statements.

   (a) Let $S \subseteq \mathbb{R}$. If $f, g\colon S \to \mathbb{R}$ are uniformly continuous functions then $f \cdot g$ is uniformly continuous on S. [2]

   Solution: Here $f \cdot g(x) = f(x)g(x)$ for every $x \in S$ is not necessarily uniformly continuous.

   For example, take $S = \mathbb{R}$ and define $f(x) = g(x) = x$ on $\mathbb{R}$. For $\varepsilon > 0$ choose $\delta = \varepsilon$ (depends only on ε) such that for any $a, x \in \mathbb{R}$ with $|x - a| < \delta$ implies that $|f(x) - f(a)| < \varepsilon$. That is, f, g are uniformly continuous.

   However, $f \cdot g$ is not uniformly continuous since there are two sequences $\{n :\ n \geq 1\}$ and $\{n + \frac{1}{n} :\ n \geq 1\}$ such that $\lim\limits_{n\to\infty} (n + \frac{1}{n} - n) = 0$ and

   $$\left|f \cdot g(n + \frac{1}{n}) - f \cdot g(n)\right| = \left|(n + \frac{1}{n})^2 - n^2\right| = 2 + \frac{1}{n^2} \geq 2,$$

   for every $n \in \mathbb{N}$.

   (b) If $h\colon S \to \mathbb{R}$ a function such that $\big\{h(x_n)\big\}$ is Cauchy sequence whenever $\{x_n\}$ is Cauchy in S then h is uniformly continuous. [2]

   Solution: This is not true. For example, the $h\colon \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x^2$ for every $x \in \mathbb{R}$ then h is not uniformly continuous as seen above. On the other hand, if $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$ then it is bounded, there is $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$ and for $\varepsilon > 0$ there exists $N_\varepsilon \in \mathbb{N}$ such that $|x_n - x_m| < \frac{\varepsilon}{M}$ for $n, m \geq N_\varepsilon$. This implies that

   $$|h(x_n) - h(x_m)| = |x_n^2 - x_m^2| = |x_n + x_m|\,|x_n - x_m| < \varepsilon, \text{ for } \ m, n \geq N_\varepsilon.$$

   This means that $\{h(x_n)\}$ is a Cauchy sequence.

   (c) If $g\colon S \to \mathbb{R}$ is a uniformly continuous function then $\big\{g(x_n)\big\}$ is a bounded sequence whenever $\{x_n\}$ is bounded in S. [2]

   Solution: Given g is uniformly continuous, for any $\varepsilon > 0$ there is a $\delta > 0$ such that

   $$|x - a| < \delta \text{ implies that } \ |g(x) - g(a)| < \varepsilon.$$

Assume that $\{g(x_n)\}$ is unbounded for some bounded sequence $\{x_n\}$ in S. By Bolzano-Weiestrass theorem there exists a subsequences $\{x_{n_k}\}$ of the sequence $\{x_n\}$ such that $\{x_{n_k}\} \to x \in \overline{S}$
and it can be chosen that $|x_{n_k} - x_{n_\ell}| < \delta$ for $k, \ell \in \mathbb{N}$ but $|f(x_{n_k}) - f(x_{n_\ell})| \geq \epsilon$ as $\{f(x_n)\}$ is unbounded. This is a contradiction. Hence g must map bounded sequence to a bounded sequence.

2. Is the series of functions $\sum\limits_{n=1}^{\infty} \frac{n^x}{e^n}$ converge uniformly on $[0, 23]$? Justify your answer. [4]

Solution: Let $f_n(x) = \frac{n^x}{e^n}$ for $x \in [0, 1]$. Suppose that $n \geq 26!$ then

$$\begin{aligned} e^n &= 1 + n + \frac{n}{2!} + \frac{n^3}{3!} + \frac{n^4}{4!} + \cdots + \frac{n^{26}}{26!} + \cdots \\ &\geq \frac{n^{26}}{26!} \\ &\geq n^{25}. \end{aligned}$$

This implies that $\frac{n^{23}}{e^n} \leq \frac{1}{n^2}$.

Now consider the sequence $M_n = \frac{1}{n^2} \geq 0$ for $n \in \mathbb{N}$ then $\sum\limits_{n=1}^{\infty} \frac{1}{n^2} < \infty$ and

$$|f_n(x)| = \frac{n^x}{e^n} \leq \frac{1}{n^2}, \text{for } n \geq 26! \text{ and every } x \in [0, 1].$$

By the Weierstrass M-test, it follows that the series $\sum\limits_{n=1}^{\infty} f_n = \sum\limits_{n=1}^{\infty} \frac{n^x}{e^n}$ converges uniformly on $[0, 23]$.

3. For each $n \in \mathbb{N}$ the function $f_n \colon [0, 1] \to \mathbb{R}$ is given by [4]

$$f_n(x) = nx(1-x)^n, \text{ for } 0 \leq x \leq 1.$$

Investigate uniform convergence of the sequence $\{f_n : n \in \mathbb{N}\}$ and justify your answer.

Solution: Clearly, $f_n(0) = 0 = f_n(1)$. Let $x \in (0, 1)$ be fixed. We choose $\delta > 0$ satisfying, $1 + \delta = \frac{1}{1-x}$ and so,

$$(1+\delta)^n \geq \frac{n(n-1)}{2}\delta^2, \text{ for } n \geq 2.$$

It follows that

$$\frac{n}{(1+\delta)^n} \leq \frac{2}{(n-1)\delta^2} \longrightarrow 0, \text{ as } n \to \infty.$$

Further, $\lim\limits_{n\to\infty} nx(1-x)^n = \lim\limits_{n\to\infty} \frac{nx}{(1+\delta)^n} = 0$. This shows that $f_n(x) \to 0$ for every $0 \leq x \leq 1$.

Since $\{f_n\} \longrightarrow 0$ pointwise, we verify that uniform convergence of $\{f_n\}$ to a zero function. Consider the sequence $\{\frac{1}{n}\}$ in $[0, 1]$ we see that

$$\lim_{n\to\infty} f_n\Big(\frac{1}{n}\Big) = \lim_{n\to\infty} (1 - \frac{1}{n})^n = \frac{1}{e}.$$

Thus there exists $N \in \mathbb{N}$ such that $|f_n(\frac{1}{n})| \geq \frac{1}{2e}$ for $n \geq N$. This implies that $\{f_n\}$ does not converge to 0 uniformly.

4. Let $f(x) = x^2$ for $0 \le x \le 1$. Compute $B_6(f), B_3(f)$ and $B_{2025}(f)$, where $B_n(f)$ denote the n^{th} Bernstein polynomial for f. [3]

Solution: The n^{th} Bernstein polynomial for a continuous function f on $[0,1]$ is given by

$$B_n(f) = \sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} f\Big(\frac{k}{n}\Big).$$

To compute the $B_n(f)$ for $f(x) = x^2,\ 0 \le x \le 1$; let us differentiate the Binomial expansion $(p+q)^n = \sum\limits_{k=0}^{n}$ with respect to p on both sides and multiply with $\frac{p}{n}$. It follows that

$$\frac{p}{n}\frac{d}{dp}\Big[(p+q)^n\Big] = \frac{p}{n}\frac{d}{dp}\sum_{k=0}^{n} p^k q^{n-k} \Leftrightarrow p(p+q)^{n-1} = \sum_{k=0}^{n}\binom{n}{k} p^k q^{n-k}\frac{k}{n}.$$

Again differentiate the above expression with respect to p on both sides and multiply with $\frac{p}{n}$, we get

$$\frac{p}{n}\,(p+q)^{n-1} + \frac{n-1}{n} p^2 (p+q)^{n-2} = \sum_{k=0}^{n}\binom{n}{k} p^k q^{n-k}\Big(\frac{k}{n}\Big)^2.$$

Put $p = x,\ q = 1-x$, we obtain

$$B_n(f) = \frac{1}{n}\,x + \frac{(n-1)}{n}\,x^2.$$

Finally, this implies that

$$B_3(f) = \frac{1}{3}(2x^2 + x),\ B_6(f) = \frac{1}{6}(5x^2 + x) \text{ and } B_{2025}(f) = \frac{1}{2025}(2024x^2 + x).$$

5. Let I be a closed and bounded interval in $\mathbb{R}$. Show that if $f\colon I \to \mathbb{R}$ is continuous then f is uniformly continuous. [3]

Solution: Assume that f is not uniformly continuous, then there exist $\varepsilon > 0$, two sequences $\{x_n\},\ \{a_n\}$ in I such that $\lim\limits_{n\to\infty}(x_n - a_n) = 0$, but

$$|f(x_n) - f(a_n)| \ge \varepsilon, \quad \text{for each } n \in \mathbb{N}$$

Since I is bounded, by Bolzano-Weierstrass theorem there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ that converges to some x in I. Moreover, $x \in I$ because I is closed. This implies that

$$|a_{n_k} - x| \le |a_{n_k} - x_{n_k}| + |x_{n_k} - x| \to 0, \text{ as } n \to \infty.$$

Since f is continuous, $\{f(x_{n_k})\} \to f(x)$ and $\{f(a_{n_k})\} \to f(x)$ then

$$|f(x_{n_k}) - f(a_{n_k})| \le |f(x_{n_k}) - f(x)| + |f(x) - f(a_{n_k})| \to 0.$$

This is a contradiction. Therefore, f is uniformly continuous.

*** ALL THE BEST ***